Roll No. | | | | | | | |

MID SEMESTER EXAMINATION [illegible] - 2025

Name of the Course: **B. Tech Petroleum Engineering** Semester: **Fourth**
Name of the Paper: **Numerical and Geostatistical methods** Paper Code: **TMA-403**
Time: **1and 1/2 Hours** Maximum Marks: **50**

Note:-

(i) Answer **all the questions** by choosing **any one of the sub questions.**

(ii) Each question carries 10 marks.

Q1	(10marks)	CO1
(a)	Solve the following system of linear equations by Gauss elimination method. $$x - y + z = 1$$ $$-3x + 2y - 3z = -6$$ $$2x - 5y + 4z = 5$$	
	OR	
(b)	Solve the following system of linear equations by Gauss- Jordan method. $$x + y + 5z = 7$$ $$2x + 10y + z = 13$$ $$10x + y + z = 12$$	
Q2	(10 marks)	CO1
(a)	Perform five iterations of the Bisection method to obtain the smallest positive root of the equation $$f(x) = x^3 - 5x + 1$$	
	OR	
(b)	Find the value of $(48)^{1/3}$ using Newton Raphson iteration method.	
Q3	(10 marks)	

(a)	A curve is drawn to pass through the points given by the following table: X: 1, 1.5, 2, 2.5, 3, 3.5, 4 y: 2, 2.4, 2.7, 2.8, 3, 2.6, 2.1 Estimate the area bounded by the curve, x-axis and the lines x=1, x=4	CO2
	OR	
(b)	Calculate an approximate value of $\int_0^{\pi} sinxdx$ by Trapezoidal rule.	
Q4	(10 marks)	
(a)	Solve the differential equation using Euler's method in six steps. $\frac{dy}{dx} = x + y,\ y(0) = 0$ choosing $h = 0.2$	CO2
	OR	
(b)	The velocities of car which starts initially from the rest (running on a straight road) at intervals of 2 minutes are given below: Time(minutes): 2, 4, 6, 8, 10, 12 Velocity (km/hr.): 22, 30, 27, 18, 7, 0 Using suitable numerical integration method to find the distance travelled by the car.	
Q5	(10 marks)	
(a)	For the equation $\frac{dy}{dx} = \frac{x^2}{2y}$, $y(0) = 2$ use suitable Runge-Kutta method to solve and find y taking $h = 0.5$.	CO2
	OR	
(b)	Solve the equation $\frac{dy}{dx} = x + 2y$ and y=1 when x=1 to find the value of y corresponding to x=2 using explicit method.	